

Wide Bay Seagrass Resilience Modeling

DRFA 21/22 Biodiversity Conservation Marine Final Report Project

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1 Introduction

Over the past few decades, various scientific studies have informed our current understanding of trends, behaviors, and flood responses of seagrass cover in Wide Bay [Preen et al., 1995, McMahon et al., 2005, York et al., 2022] as well as Australia more widely [Kilminster et al., 2015, Gilby et al., 2018, O'Brien et al., 2018]. Remote sensing has offered new capabilities for seagrass cover mapping [Roelfsema and Phinn, 2009], while innovative statistical approaches have offered novel methods for studying resilience at site-specific scales [Gilby et al., 2023]. Our work adds to this body of literature by applying spatially explicit statistical models [Cressie, 1993] to study the drivers, resilience, and recovery of total percent seagrass cover in Wide Bay and Moreton Bay [Dumelle and Peterson, 2026]. Specifically, we provide and contextualize answers to three specific research questions, applicable to Wide Bay:

1. Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or ambient conditions?
2. Did seagrass cover improve after the flood event(s)? If so, how and where?
3. Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded? If so, what factors contribute to seagrass resilience?

2 Seagrass Cover Model

2.1 Data

The seagrass cover dataset was provided by EcoFutures Consulting. It included nearly 4000 seagrass cover observations in Wide Bay across several financial years (1998-1999, 1999-2000, 2000-2001, 2001-2002, 2021-2022, 2022-2023, and 2023-2024). The data from 1998-2002 were omitted as they were distant temporally and did not have corresponding eReefs water-quality data [Steven et al., 2019]. Thus we proceed with a subset of 2663 observations for modeling. Of the 2663 observations, 1073 were in 2021-2022, 524 were in 2022-2023, and 1066 were in 2023-2024 (Table 1). There were four seagrass species found in the dataset: *Halophila decipiens* (HD), *Halophila spinulosa* (HS), *Halophila ovalis* (HO), *Halodule uninervis* / *Zostera muelleri* (ZMHU).

Seagrass cover is a percentage that ranges from 0 (no seagrass) to 100 (complete seagrass cover). Figure 1 shows total cover by financial year throughout the Bay. Total cover tends to be highest near the eastern shoreline of the Bay and lowest in the middle of the Bay. Note that 2022-2023 contained the fewest number of samples, lacking significant samples in the Great Sandy Strait compared to the other years.

EcoFutures Consulting used eReefs data [Steven et al., 2019] to generate a broad suite of ambient and flood-related turbidity, salinity, and temperature variables at biologically and

Table 1: The sample size and percent of total sample size for total percent seagrass cover observations in Wide Bay, by financial year.

Financial Year	Sample Size	Percent of Total
2021-2022	1073.00	40.3%
2022-2023	524.00	19.7%
2023-2024	1066.00	40.0%
Total	2663.00	100.0%

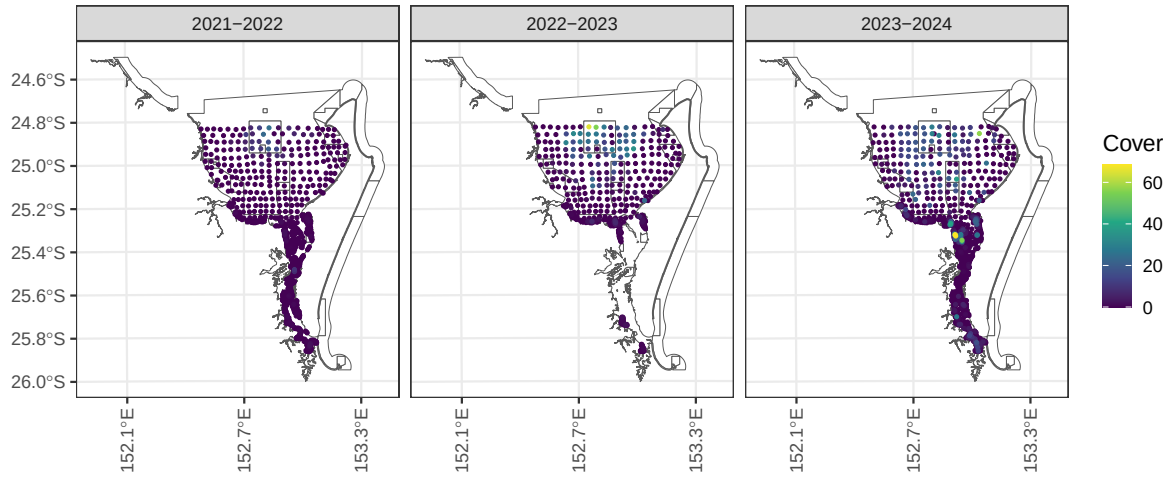


Figure 1: Total seagrass cover in Wide Bay by financial year.

ecologically relevant temporal windows (Table 2). Ambient water quality variables were based on the mean, minimum, and maximum values for three periods: 7, 30, and 90 days prior to sampling. Ambient variables were also calculated representing the frequency of days water quality thresholds were exceeded (above or below) and the maximum spell (i.e., duration) of days those thresholds were continuously exceeded during the time periods. Similar water quality variables representing flood-event conditions were also calculated for the two separate 30-day flood periods.

Additional covariates were also provided by EcoFutures Consulting, which are used to represent geographic features, depth, and algae cover, as well as the years since the floods and the seagrass growing season. These variables have the following structure:

- Sediment Type has six categorical levels: Sand, Sandy Mud, Muddy Sand, Mud, and Rock.
- Tidal Range has three categorical levels: Shallow Subtidal, Intertidal, and Deep Subtidal.
- Years Since Flood One is numeric ranging from 0.096 (May, 2022) to 1.65 (November, 2023).
- Years Since Flood Two is numeric ranging from 0 (May, 2022) to 1.65 (November, 2023).
- Algae Presence has two categorical levels: Algae present and algae not present.
- Algae Cover is numeric ranging from 0 (no algae) to 100 (entire area covered in algae).
- Growing Season has two categorical two levels: Growing (September - December) and Not Growing (January - August).
- Zone class has six categorical levels: Central, East, Great Sandy Strait, Hervey Bay, Southern, and Southwest (Figure 2).

2.2 Exploratory Analyses

Extensive exploratory analyses were undertaken to 1) understand the data structure, 2) visualize spatial and temporal patterns in the data, 3) identify outliers/errors, 4) examine correlations between potential covariates and the seagrass cover, as well as multicollinearity among covariates, 5) compare the fit of various models, and 6) evaluate model assumptions.

Results revealed substantial collinearity among ambient and flood-related water-quality variables, as expected given their nested temporal windows and threshold values. We retained only one variable from each correlated group. We prioritized including sediment type, tidal range, turbidity, temperature, and time since floods, given a goal was to study the extent to which these variables were related to seagrass cover. While we had data for time since both floods, we removed time since the second flood because these two variables were highly collinear. The majority of the data were collected after the second flood, in which case the variables are redundant. Rather than trying to separate water quality effects from each of the two floods, we averaged these effects across both floods to create a single turbidity metric capturing flood-related impacts. We removed other variables showing no meaningful association with seagrass cover or where the relationship lacked biological plausibility. This included

Table 2: Wide Bay water quality variables considered in the exploratory analyses, including the variable name and measurement unit (in parentheses), the type of water-quality variable (Ambient, Flood), the time period in days, statistic calculated, and the threshold value used (if any).

Variable	Type	Period	Statistic	Threshold
Turbidity (NTU)	Ambient	30, 7, 90	Mean	
Turbidity (NTU)	Ambient	30, 7, 90	Frequency	> 3
Turbidity (NTU)	Ambient	30, 7, 90	Frequency	> 5
Turbidity (NTU)	Ambient	30, 7, 90	Frequency	> 6
Salinity (PSU)	Ambient	30, 7, 90	Frequency	< 20
Salinity (PSU)	Ambient	30, 7, 90	Frequency	< 25
Salinity (PSU)	Ambient	30, 7, 90	Frequency	< 30
Temperature (deg C)	Ambient	30, 7, 90	Mean	
Turbidity (NTU)	Flood	30	Mean	
Turbidity (NTU)	Flood	30	Max	
Turbidity (NTU)	Flood	30	Frequency	> 3
Turbidity (NTU)	Flood	30	Frequency	> 5
Turbidity (NTU)	Flood	30	Frequency	> 6
Turbidity (NTU)	Flood	30	Duration	> 3
Turbidity (NTU)	Flood	30	Duration	> 5
Turbidity (NTU)	Flood	30	Duration	> 6
Salinity (PSU)	Flood	30	Min	
Salinity (PSU)	Flood	30	Mean	
Salinity (PSU)	Flood	30	Frequency	< 20
Salinity (PSU)	Flood	30	Frequency	< 25
Salinity (PSU)	Flood	30	Frequency	< 30
Salinity (PSU)	Flood	30	Duration	< 20
Salinity (PSU)	Flood	30	Duration	< 25
Salinity (PSU)	Flood	30	Duration	< 30

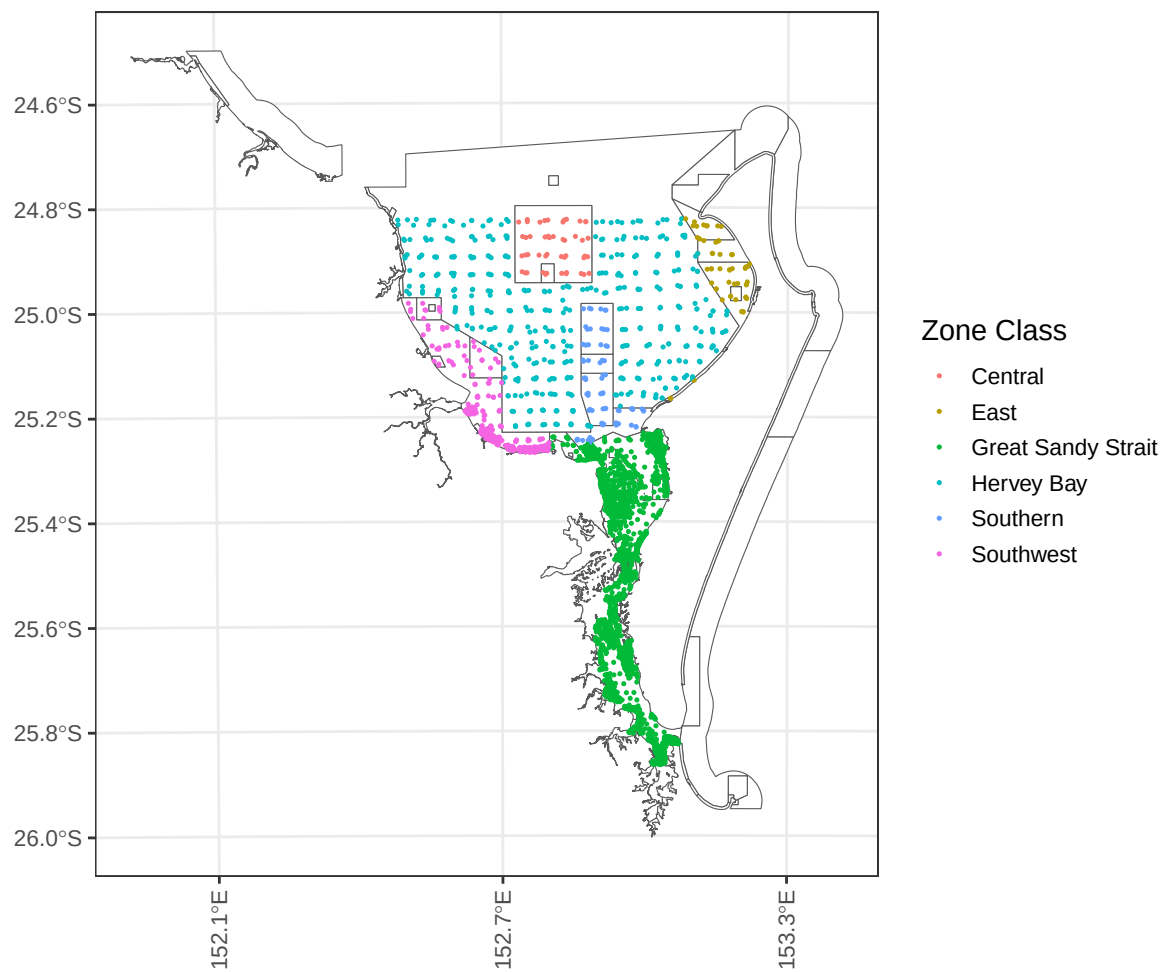


Figure 2: Observations and their zone classes in Wide Bay.

the salinity variables, which had little variability and did not exhibit strong relationships with seagrass cover. Geographic variables exhibited less collinearity than water quality variables; however, we removed the algae variables due to weak associations with seagrass cover after accounting for other variables in the model. Growing season was excluded because a simple monthly proxy inadequately captures seasonal seagrass growth dynamics.

The final set of potential covariates considered in the modelling included mean turbidity in the 90 days prior to sampling, mean turbidity during the flood event, mean temperature in the 90 days prior to sampling, sediment type, tidal range, years since flood one, and EHMP subzone.

2.3 Statistical Modeling

We considered multiple modeling approaches in this study, which each accounted for the data distribution, nonlinear relationships between seagrass cover and covariates, and spatial dependence inherent in biological data collected through space and time to different degrees (Appendix 1). We used the full set of potential covariates to fit a nonspatial linear model, spatial linear model (i.e., spatial model), spatial beta regression model, spatial gamma regression model, and a random forest model, comparing the model fit of each. We also considered a spatial logistic regression model, treating seagrass as presence/absence data; this model had similar drivers as the total seagrass models and so we did not consider it further.

There were several reasons for fitting these initial models. The nonspatial linear model provided a useful baseline to compare the spatial linear model against, elucidating the impact of spatial dependence on model fit. The spatial beta regression model is a generalized linear model used to characterize proportion data ranging between zero and one. To adhere to this restriction, total seagrass cover was 1) scaled from a percentage to a proportion and 2) if exactly zero or one, set to 0.01 or 0.99, respectively. The spatial gamma regression model is a generalized linear model used to characterize skewed positive data. To adhere to this restriction, a value of 0.01 was added to seagrass cover observations that were equal to zero. Finally, the random forest model provided a useful assessment of a machine learning approach, which requires fewer distributional assumptions than the aforementioned statistical approaches and more naturally describes nonlinear relationships between variables (Appendix 1). We also characterized random forest effects using partial dependence plots [Greenwell, 2017], which isolate the marginal contribution of a variable while holding other variables at their observed values. Out-of-sample predictions were generated for the four statistical models using leave-one-out cross validation, while out-of-bag validation was used for the random forest model. These out-of-sample predictions were then used to calculate the mean bias (MBias) and root-mean-square-prediction error (RMSPE) for each model.

The results of this initial model comparison showed that the random forest had the lowest RMSPE, followed by the spatial linear model, generalized linear models, and the nonspatial linear model (Table 3). The nonspatial and spatial linear models also had considerably lower

Table 3: Wide Bay out of sample prediction metrics for the various approaches. Mean bias (MBias) tended to be small relative to root-mean-squared-prediction error (RMSPE). For LM, SPLM, SPGLM-Beta, SPGLM-Gamma, leave-one-out cross validation was used. For RF, out of bag validation was used.

Approach	MBias	RMSPE
Random Forest	-0.029	4.537
Spatial Linear Model	-0.003	4.700
Spatial GLM - Gamma	1.177	5.116
Spatial GLM - Beta	-0.159	5.306
Linear Model	-0.000	5.452

mean bias values than the random forest model, while the two generalized linear models produced unacceptably biased predictions. Overall, all five approaches generally identified similar relationships between seagrass cover and the covariates, in the directions expected (when there were significant effects). There were some minor differences in the effect of turbidity among the modeling approaches, which we briefly note later. Random forest partial dependence plots [Greenwell, 2017] suggested that linear effects of the model variables were generally reasonable. Given these results and the ease of interpretability of the spatial linear model, we chose to use a spatial linear modeling framework for the Wide Bay seagrass cover model.

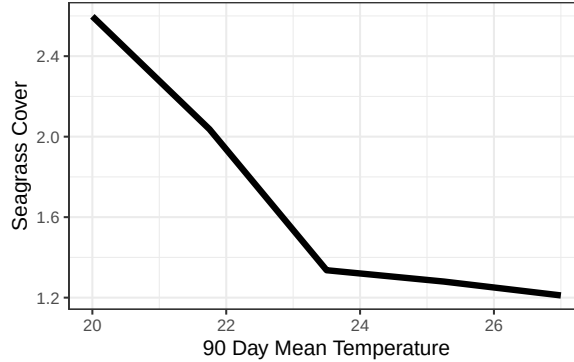


Figure 3: Random forest partial dependence for total seagrass percent cover as a function of 90 day mean temperature.

A suite of spatial linear models (Appendix 1, Equation 1), with exponential spatial covariance functions and a random effect for zone class, were fit using the `splm()` function in `splmodel` [Dumelle et al. [2023]]. Parameters were estimated using restricted maximum likelihood. Parameters were estimated using restricted maximum likelihood and models primarily compared model using leave-one-out cross validation fit statistics; though AIC or BIC could also be used if parameters were estimated using maximum likelihood. In addition, the relationships between seagrass cover and covariates needed to be plausible.

Table 4: Covariates in the final Wide Bay seagrass cover model, their type (numeric or categorical), and if categorical, the number of unique levels.

Variable	Type	No. Levels
90 Day Temperature	Numeric	NA
90 Day Turbidity	Numeric	NA
Mean Flood Turbidity/Years Since Flood One ²	Numeric	NA
Sediment Type	Categorical	5
Tidal Range	Categorical	3

Table 5: Fixed effects table of covariates in the Wide Bay water quality and geographic model.

Term	Estimate	z	p.value
Intercept (Mud/Deep Subtidal)	14.405	5.111	0.000
Intertidal	0.419	0.969	0.333
Shallow Subtidal	1.652	1.955	0.051
Muddy Sand	-1.060	-1.822	0.068
Rock	-1.264	-1.282	0.200
Sand	-0.685	-1.195	0.232
Sandy Mud	-1.492	-2.700	0.007
90 Day Temperature	-0.502	-7.411	0.000
90 Day Turbidity	0.844	1.371	0.170
Mean Flood Turbidity/Years Since Flood One ²	-0.000	-0.050	0.960

The final total seagrass cover model for Wide Bay included the following covariates: tidal range, sediment type, 90-day mean temperature (prior to sampling), 90 day turbidity (prior to sampling), and mean turbidity during flood period one scaled by time since the flood squared (Table 4). Mean turbidity during flood period one was scaled so that its effect decays with time since the flood. Intuitively, this means that as the time since flood one increases, the impact of turbidity during the flood event decreases. We also considered scaling by time since flood one (not squared), which produced similar results. This modeling approach is similar to the Moreton Bay modeling approach [Dumelle and Peterson, 2026], as similar research questions were asked for each region.

Table 5 shows the fixed effects table from the fitted model. The reference group is the muddy and deep subtidal group, and the remaining sediment type and tidal range terms represent deviations from these baselines. For the numeric variables, temperature had a small p -value (p -value < 0.01) while the turbidity metrics had large p -values (p -values > 0.1). While we can assess the overall significance of the numeric variables Table 5, we must perform an analysis of variance (ANOVA) to assess the overall significance of variables with many levels. Table 6 shows these ANOVA results for the sediment type and tidal range variables; sediment type had

Table 6: ANOVA table of covariates for the categorical variables in the Wide Bay water quality and geographic model.

Effects	df	Chi-sq	p.value
Sediment Type	4	16.638	0.002
Tidal Range	2	3.870	0.144

a small p -value (p -value < 0.01) while tidal range had a larger p -value (p -value > 0.1). Together, these results indicate sediment type and temperature had significant effects on seagrass cover, while there was not substantial evidence the other variables did, at least in these data. Here are some relevant takeaways from the model fit for the sediment type and temperature variables:

- Sediment Type: The estimated (marginal) average seagrass cover is largest in mud (4.3%), followed by sand (3.6%), muddy sand (3.2%), rock (3.0%), and sandy mud (2.8%).
- 90 Day Temperature: A one-degree (C) increase in 90 day temperature is associated with a decrease in average total cover by 0.5% points.

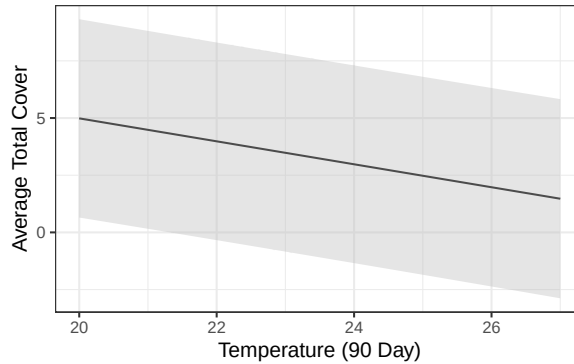


Figure 4: Average total cover as a function of 90 day mean temperature. Estimates (and confidence intervals) were evaluated at the means of all other numeric variables, and for the intertidal range and Sand sediment types.

We examined model diagnostics to better understand the overall model fit. First, we partitioned the model into different components of variability (Table 7) and found that 5.4% is attributable to the covariates (a quantity sometimes referred to as the Pseudo R-squared), 54.2% to the spatial random error, 5.8% to the zone class, and the remaining variability is independent random error. Second, we examined the decay behaviour of spatial autocorrelation. The estimated exponential spatial range (ϕ) is 14757 meters, which suggests two total cover observations are approximately (spatially) uncorrelated at a distance of $3\phi \approx 45\text{km}$, or forty-five kilometers. Third, we analyzed the standardized model residuals, which have been decorrelated, to assess model assumptions and identify influential observations that had a sizable impact on model fit. Generally, the standardized residuals are centered around zero,

Table 7: Variance components in the Wide Bay seagrass model.

Variance Component	Percent (of Total Variability)	Parameter Value
Covariates	5.4%	
Spatial Effect	54.2%	28.8
Independent Error	34.6%	18.4
Zone Class	5.8%	14757.0

though there is a slight right skew suggesting some observed total cover values are larger than expected (according to the model). Observations are referred to as “influential” if they highly influence model fit. Influence is often assessed using a Cook’s distance threshold value of one [Cook and Weisberg, 1982]. No observations exceed this threshold.

All five modeling approaches found significant relationships between sediment type and temperature. The nonspatial linear model and spatial beta regression model found significant, negative relationships between turbidity and total cover overall, while the random forest ranked turbidity as important metrics. All but the spatial linear model and random forest found significant relationships among tidal ranges, with intertidal or deep subtidal having the lowest average total cover. For the random forest, tidal range was the least important variable.

2.4 Block Kriging

A tremendous benefit of using the spatial linear model is that block Kriging [Ver Hoef, 2002, Cressie, 2006] can be used to predict the mean total seagrass cover bay-wide for any combination of covariates and geographic regions, even if these variables were not directly used in the model (see Appendix 1 for additional details). Block Kriging was undertaken, by financial year, for zone class, Marine Park zone, tidal range and for the whole of Wide Bay.

Figure 5, Figure 6, and Figure 7 show that seagrass cover has generally improved since 2021-2022 across all zone classes, conservation zones, and tidal ranges. Although there were notable declines observed in Marine National Park Zones between 2022-2023 and 2023-2024 (Figure 6). When we examine recovery for the whole of Wide Bay (Figure 8), the model suggests that seagrass cover has improved since 2021-2022, but the same declines are observed between 2022-2023 and 2023-2024.

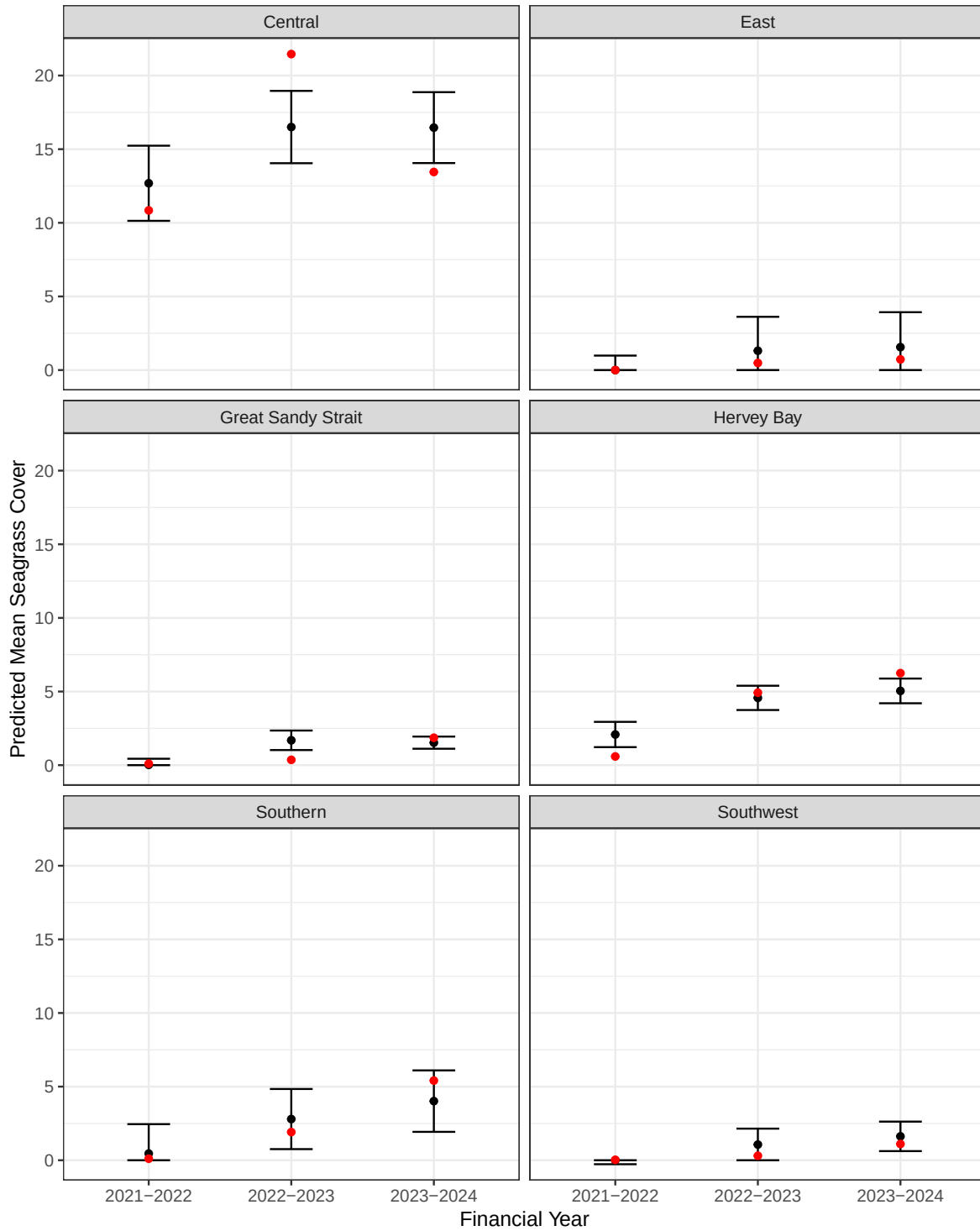


Figure 5: Average seagrass cover predictions by zone class. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

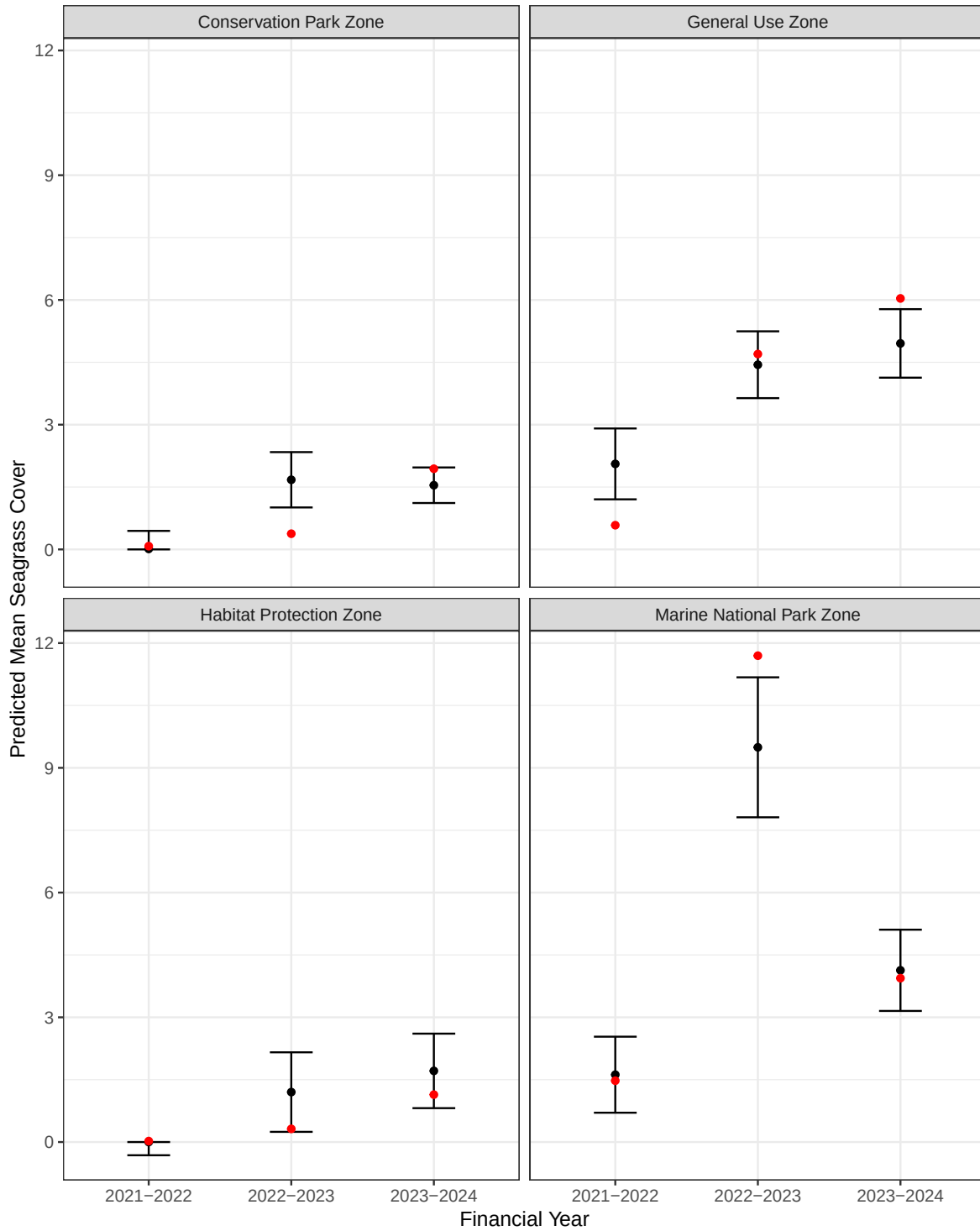


Figure 6: Average seagrass cover predictions by conservation zone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

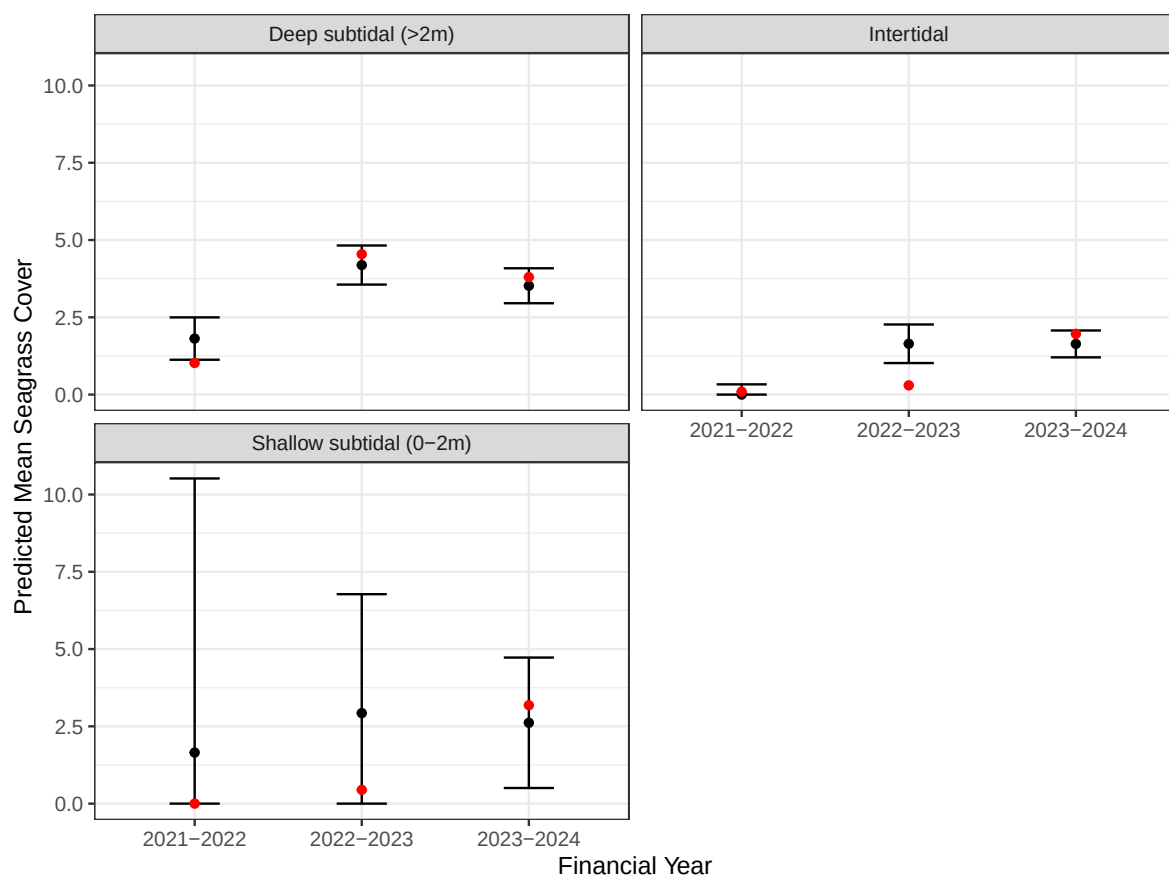


Figure 7: Average seagrass cover predictions by tidal range. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

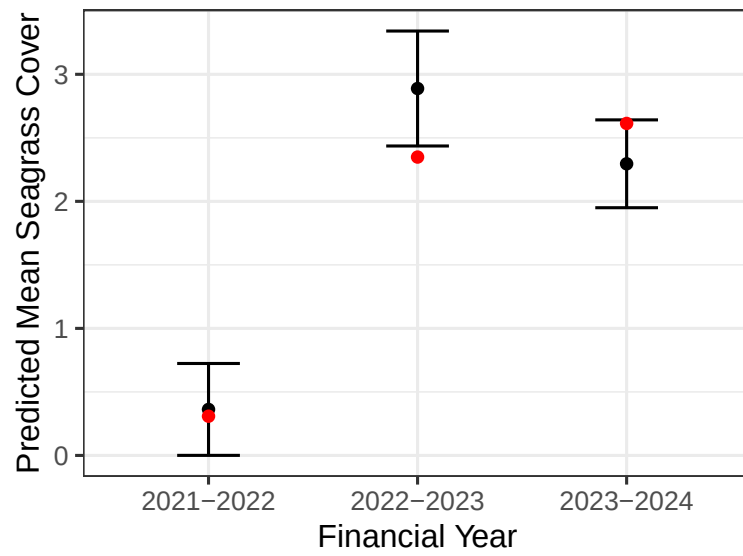


Figure 8: Average seagrass cover predictions in Wide Bay. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

3 Resilience Modeling

The seagrass cover model quantifies bay-wide effects of influential drivers on seagrass cover and enables assessment of recovery trajectories at management-relevant scales. However, it does not quantify site-specific resilience. Gilby et al. [2023] propose identifying hotspots (i.e., brightspots, sites performing better than expected) and coldspots (sites performing worse than expected) based on model residuals exceeding confidence interval bounds. They argue that hotspots and coldspots offer significant utility for understanding drivers of site-specific resilience and informing management decisions. Gilby et al. [2023] primarily focused on specific hotspots, linking environmental variables to those sites. We build on that work by using the seagrass model residuals as a proxy for site-specific resilience, and then characterizing the most important drivers of that resilience.

3.1 Data

We used the raw seagrass model residuals (Figure 9) as the response variable in a second spatial linear model used to study site-specific resilience in Wide Bay. Model covariates included variables describing seagrass richness and dominant species, and bathymetry, as well as distance to the nearest coral, seagrass, mangrove, or gastropod habitat, and the total area of coral, seagrass, mangrove, or gastropod habitat within 500 m of the observation point (Table 8). The Zone class was again included as a random effect. Because we were interested in hypothesis testing rather than parsimony, we included all potential covariates as fixed effects in the model. Potential covariates have the following structure:

- Coral Habitat Total Area is numeric from 0 to 367,906 meters squared (msq).
- Seagrass Habitat Total Area is numeric from 0 to 773,161 msq.
- Mangrove Habitat Total Area is numeric from 0 to 241,057 msq.
- Distance to Coral Habitat is numeric from 0 to 30,563 m.
- Distance to Seagrass Habitat is numeric from 0 to 7,049 m.
- Distance to Mangrove Habitat is numeric from 0 to 38,867 m.
- Distance to Gastropod Habitat is numeric from 0 to 121,609 m.
- Habitat Count is numeric (integer) from 0 to 3 unique habitats (coral, seagrass, mangrove).
- Species Richness is numeric (integer) from 0 to 4 unique species (possible species: “Absent”, “HD”, “HO”, “HS”, and “ZMHU”).
- Dominant Species represents the dominant seagrass species and has five levels: “Absent”, “HD”, “HO”, “HS”, and “ZMHU”.
- Bathymetry Mean Depth is the GBR mean depth within 500 m of the seagrass observation point, excluding terrestrial areas, and is numeric from -30.58 m to 0.25 m.
- Bathymetry Mean Curvature is the GRB mean curvature within 500 m of the seagrass observation point, excluding terrestrial areas, and is numeric from -0.001 to 0.001.

- Bathymetry Mean Slope is the GBR mean slope within 500 m of the seagrass observation point, excluding terrestrial areas, and is numeric from 0.004 degrees to 3.93 degrees.
- Zone class has six categorical levels: Central, East, Great Sandy Strait, Hervey Bay, Southern, and Southwest.

We expected there to be nonlinear relationships between the response and covariates and we accounted for them using B-spline basis functions [de Boor, 1978, Eilers and Marx, 1996]. The degree of these functions was informed by partial dependence plots derived from a random forest model fitted to the seagrass model residuals (Table 8). We did not use the “Gastropod Habitat Total Area” variable because there was no gastropod habitat overlap in the Wide Bay data.

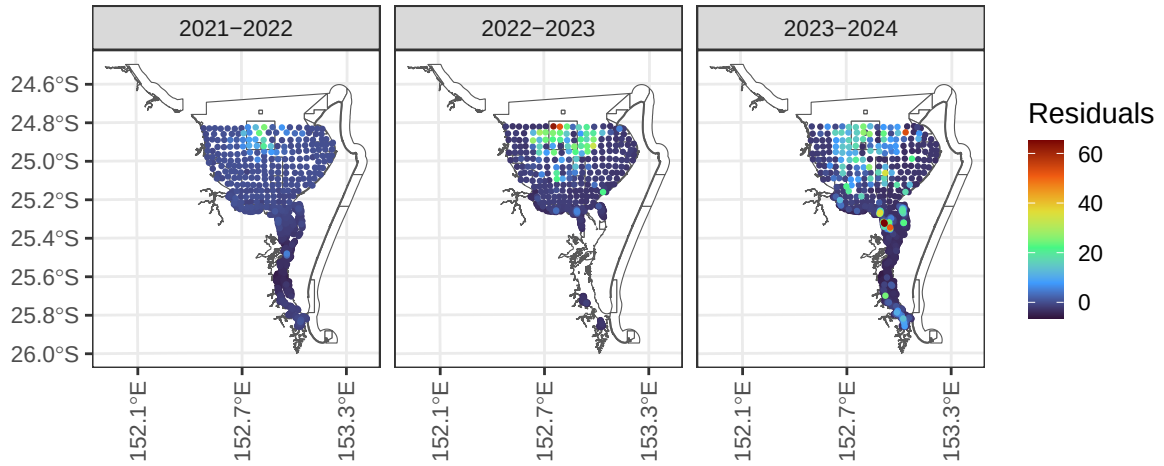


Figure 9: Seagrass model residuals for resilience analysis in Wide Bay, by financial year.

Table 8: Covariates included in the resilience model, including whether they were a fixed or random effect, and the nonlinearity degree of the B-splines basis function.

Variable	Effect	Degree
Coral Habitat Total Area	Fixed	2
Seagrass Habitat Total Area	Fixed	2
Mangrove Habitat Total Area	Fixed	2
Distance to Coral Habitat	Fixed	2
Distance to Seagrass Habitat	Fixed	2
Distance to Mangrove Habitat	Fixed	2
Distance to Gastropod Habitat	Fixed	2
Species Richness	Fixed	1
Habitat Count	Fixed	2
Dominant Species	Fixed	NA
Bathymetry Mean Depth	Fixed	2
Bathymetry Mean Curvature	Fixed	2
Bathymetry Mean Slope	Fixed	2
Zone Class	Random	NA

3.2 Statistical Modeling

Table 9 shows the fixed effects table from the fitted model. The reference group is the absent dominant species, and the remaining dominant species terms represent deviations from this baseline. Species richness, the only numeric variable with a single level, had a large test statistic and small p -value (p -value < 0.01). The other numeric variables were modeled using a B-spline basis of degree two to capture nonlinearities. While we can assess the overall significance of species richness in Table 9, we must perform an analysis of variance (ANOVA) to assess the overall significance the remaining variables with multiple terms. Table 10 shows these ANOVA results.

We found strong evidence (p -value < 0.001) that species richness, dominant species, and distance to coral habitat are associated with resilience:

- Species Richness: An increase of species richness by one species was associated with an average increase in resilience by approximately five points (Figure 10).
- Dominant Species: HS are the most resilient, followed by HD, HO, and ZMHU.
- Distance to Nearest Coral Habitat: From 0 to around 10000 m, there is a decreasing trend with resilience (i.e., increasing distance decreases resilience). Upwards of 10000 m, there is an increasing trend with resilience (i.e., increasing distance increases resilience).

We found little evidence (p -value > 0.1) the remaining variables were associated with resilience (after controlling for other modeled variables).

Table 9: Fixed effects table of covariates in the Wide Bay resilience model.

Term	Estimate	z	p.value
Intercept (Absent)	-1.278	-0.416	0.677
HD	-1.329	-1.936	0.053
HO	-1.659	-2.160	0.031
HS	1.427	1.893	0.058
ZMHU	-2.274	-4.833	0.000
Species Richness	5.245	15.449	0.000
Coral Area Deg1	0.757	0.305	0.760
Coral Area Deg2	-0.167	-0.057	0.954
Seagrass Area Deg1	-0.745	-0.967	0.334
Seagrass Area Deg2	0.408	0.774	0.439
Mangrove Area Deg1	0.973	0.498	0.619
Mangrove Area Deg2	-2.951	-1.058	0.290
Habitat Count Deg1	-0.146	-0.166	0.868
Habitat Count Deg2	-0.143	-0.169	0.866
Coral Distance Deg1	-5.142	-4.164	0.000
Coral Distance Deg2	11.490	7.074	0.000
Seagrass Distance Deg1	-0.996	-0.485	0.628
Seagrass Distance Deg2	-0.922	-0.440	0.660
Mangrove Distance Deg1	1.532	0.885	0.376
Mangrove Distance Deg2	-2.174	-1.279	0.201
Gastropod Distance Deg1	-1.897	-0.919	0.358
Gastropod Distance Deg2	-2.619	-1.747	0.081
Bathymetry Mean Depth Deg1	1.630	0.669	0.503
Bathymetry Mean Depth Deg2	1.192	0.621	0.535
Bathymetry Mean Slope Deg1	-1.434	-0.894	0.371
Bathymetry Mean Slope Deg2	0.920	0.264	0.792
Bathymetry Mean Curvature Deg1	0.465	0.169	0.866
Bathymetry Mean Curvature Deg2	-0.014	-0.007	0.994

Table 10: ANOVA table of covariates in the Wide Bay resilience model.

Effects	df	Chi-sq	p.value
Dominant Species	4	68.011	0.000
Distance to Coral Habitat	2	51.788	0.000
Distance to Gastropod Habitat	2	3.089	0.213
Distance to Mangrove Habitat	2	2.866	0.239
Seagrass Habitat Total Area	2	2.752	0.253
Mangrove Habitat Total Area	2	1.163	0.559
Bathymetry Mean Slope	2	0.810	0.667
Distance to Seagrass Habitat	2	0.536	0.765
Bathymetry Mean Depth	2	0.466	0.792
Bathymetry Mean Curvature	2	0.119	0.942
Coral Habitat Total Area	2	0.096	0.953
Habitat Count	2	0.059	0.971

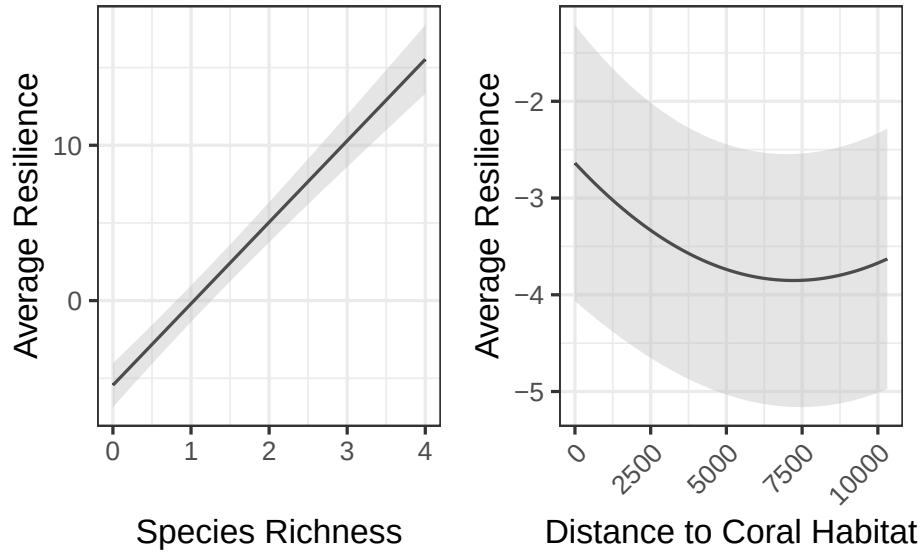


Figure 10: Average species richness (left) and distance to coral habitat (right, for distances less than 10000 m) effects on resilience. Estimates (and confidence intervals) were evaluated at the means of all other numeric variables and for the ZMHU species in the Central zone.

As mentioned previously, we also fit a random forest model to the seagrass model residuals. There were similarities and differences between the spatial linear model and random forest. Notably, for the random forest:

- Species richness was most important, with a general increasing trend in resilience with richness.
- Distance to mangrove habitat was second-most important, with a general increasing trend in resilience with distance from mangroves.
- Dominant species was third-most important, with HS being the most resilient followed by HO, ZMHU, and HD.
- Bathymetry mean depth was fourth-most important, with a general increasing trend in resilience with depth.
- Distance to gastropod habitat was fifth-most important, with a general decreasing trend in resilience with distance.
- Distance to coral habitat was sixth-most important, with a similar quadratic trend (Figure 11) as in the spatial linear model.
- Seagrass area was seventh-most important, with a general increasing trend in resilience with area.
- The remaining variables (seagrass area, bathymetry slope, bathymetry curvature, zone class, seagrass distance, habitat count, mangrove area, and coral area) had notable drop offs in performance.

The consistency between the two resilience models' variable importance rankings for species richness, dominant species, bathymetry mean depth, and distance to both gastropod and coral habitats reinforces the spatial linear model's identification of the most important drivers of resilience. However, the random forest identifies distance to mangrove habitat as the second most important variable, which is a questionable finding given the direction of the relationship. This counterintuitive result is likely a spurious correlation or an artifact of the random forest model's inability to account for spatial dependence in the data, which can affect variable importance measures [Fox et al., 2020].

4 Limitations

While the spatial linear model was useful for characterizing broad patterns in seagrass cover related to tidal range, sediment type, temperature, and turbidity and site-specific patterns capturing resilience, all while accounting for spatial dependence, there are some limitations. First and foremost, seagrass cover data were not available in the years prior to the first flood. Without a baseline, it is impossible to quantify whether seagrass cover has recovered to pre-flood levels. Second, the Wide Bay data are heavily skewed, with over 77% of observations containing no seagrass. This extreme skew compromised model fit across all the approaches we explored. We selected the spatial linear model because it naturally describes spatial dependence, offers interpretability, and enables block Kriging predictions. However, the model

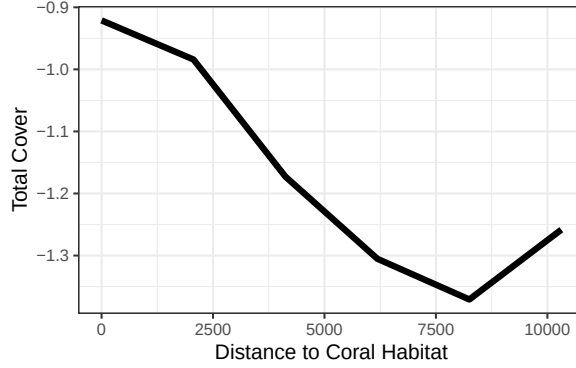


Figure 11: Random forest partial dependence for total cover as a function of distance to coral habitat (for distances less than 10000 m).

assumptions are less likely to hold given the sparse nonzero observations, which undermines the reliability of model inferences. Third, the eReefs water quality covariates are based on modeled data that are not directly observed in the field, potentially propagating modeling error. Fourth, there are many correlated water-quality variables, making it difficult (for any model) to isolate the effects of each component of water quality. Fifth, the lack of a covariate describing mean salinity in the 90 days prior to sampling makes it challenging to directly study the effect of mean salinity during the flood period. Sixth, covariates representing local-scale drivers of seagrass cover (e.g., light attenuation, biological interactions, etc.) were not available, which might explain some additional variability in the response through space and time. Finally, the location of sample sites was not generated randomly and could be subject to conscious or unconscious selection bias.

5 Revisiting the Research Questions

Research Question 1: Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?

We identified temperature as the primary water quality variable in the seagrass cover model, with a one-degree increase in 90-day mean temperature associated with a 0.5 percentage point decrease in cover. This ambient temperature signal was statistically significant and ecologically interpretable, as elevated temperatures may compound the effects of other stressors on seagrass. In contrast, neither ambient turbidity nor flood turbidity showed strong statistical evidence of association with total cover after accounting for other variables. This could result in part due to the absence of pre-flood baseline data, which made it difficult to isolate the contribution of flood-driven turbidity from background conditions. It could also be in part due to the challenges associated with analyzing highly skewed data with many zeros. Across tidal ranges,

estimated average total cover was modest throughout, and sediment type was a significant predictor, whereby muddy substrates supported slightly higher average seagrass cover than sand, sandy mud, or rock.

Research Question 2: Did seagrass cover improve after the flood event(s)? If so, how and where?

Total seagrass cover has generally improved throughout Wide Bay between financial years 2021-2022 and 2023-2024, suggesting the system may be recovering from the floods. This improvement was seen across zone classes, conservation zone types, tidal ranges, and bay-wide. However, a decline in cover was observed in the Marine National Park Zone between 2022-2023 and 2023-2024 and this likely contributes to the slight decline observed for Wide Bay as a whole in the same years. These findings should be interpreted with care. The absence of pre-flood baseline data means it is impossible to assess whether the observed cover levels represent recovery relative to a pre-disturbance state or simply reflect the current trajectory of the ecosystem.

Research Question 3: Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded? What factors contribute to seagrass resilience?

We identified several site-specific indicators of seagrass resilience in Wide Bay. Species richness and dominant species were the strongest indicators of resilience, with each additional species associated with an average increase in resilience of approximately five points. Sites dominated by HS tend to be most resilient, followed by HD, HO, and ZMHU. Distance to coral habitat was also a strong resilience indicator, exhibiting a nonlinear pattern. Resilience declined with increasing distance from coral up to approximately 10000 m, then increased beyond that threshold, a pattern consistent with coral and seagrass occupying different but adjacent positions along Wide Bay's water quality and substrate gradient. These findings were broadly corroborated by the random forest, which ranked species richness as the most important predictor, dominant species, distance to mangrove habitat, bathymetry mean depth, distance to gastropod habitat, distance to coral habitat, and seagrass area. Together, these results suggest that management actions protecting structurally diverse, species-rich meadows are likely to be most effective at improving resilience, while sites with low species richness (or no seagrass) may require more active intervention to support recovery.

6 Summary

We applied a rigorous, spatially explicit statistical framework to characterize the drivers, recovery, and resilience of seagrass cover in Wide Bay across a three-year monitoring period that included the two separate Wide Bay flood events. The analytical approach taken here is well-suited to the structure and scale of the data, and the findings have direct implications for adaptive management of the bay's seagrass ecosystem.

The use of spatial linear models as the primary analytical framework is well-justified from both statistical and ecological perspectives. Total percent seagrass cover observations collected across the bay are inherently spatially structured, as nearby sites share similar environmental conditions, water quality and disturbance histories, as well as ecological communities. Models that ignore this dependence may well provide misleading results and inference. By explicitly modeling spatial dependence via a spatial covariance function, the analyses presented here produce more reliable estimates of covariate effects and more honest characterizations of uncertainty than a conventional non-spatial model provides. These models are typically also more interpretable and characterize uncertainty more effectively than machine learning models. The inclusion of zone class as a random effect further accounts for structured variation among management-relevant regions of the bay, ensuring that covariate estimates reflect within-zone patterns rather than being confounded by between-zone differences in baseline cover. The block Kriging helps understand bay-wide trends and can immediately characterize bay-wide predictions on the original total seagrass cover scale (0-100), rather than a transformed scale required for generalized linear models; providing a flexible framework for scenario modeling under future flood conditions and climate scenarios.

The use of a two-stage modeling framework – first modeling total seagrass cover as a function of water quality and geographic drivers, then modeling the seagrass model residuals as a function of site-level resilience covariates – is a structured approach to separating bay-wide environmental patterns from site-specific ecological resilience capacity. This builds directly on the framework proposed by [Gilby et al. \[2023\]](#) to help understand drivers of average resilience and provide useful insight needed to inform broader conservation strategies. The robustness of the findings is further supported by the general consistency of results across multiple modeling approaches (e.g., spatial beta regression model, random forest, etc.), which provides evidence that the relationships identified are genuine rather than resulting solely from specific modeling frameworks.

Overall, the results described in this report provide a quantitative, spatially structured foundation for adaptive decision making. Together, the identification of important water quality and geographic variables, bay-wide predictions via block Kriging, and the characterization of site-level resilience indicators provide data-driven evidence that can inform temporally targeted, context-specific interventions throughout the bay.

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Appendix 1. Spatial Linear Models and Random Forest Models

The spatially explicit statistical models (i.e., spatial models) we used to answer these research questions are related to commonly used linear and generalized linear models but incorporate *spatial dependence*. Spatial dependence measures how similar two sites are based on their distance. Formally, we can build spatial dependence directly into spatial models by leveraging Tobler’s First Law of Geography [Tobler, 1970], which states succinctly that “nearby sites tend to be more similar than distant sites”. Not only is this intuitive from a “first principles” perspective, but spatial models also provide many advantages for modeling spatial data compared to models that assume independence among observations (i.e., nonspatial models). First, nonspatial models tend to underestimate uncertainty, yielding covariate standard errors and p-values for covariates that are too small and corresponding confidence intervals that are too narrow. Second, spatial models improve prediction at unsampled locations by borrowing strength from nearby observations, yielding more precise predictions and corresponding standard errors that vary by location. Third, spatial models decompose model error into separate spatial and nonspatial components, informing residual and resilience structure in ways that nonspatial models cannot. Fourth, spatial models can provide ecologically meaningful insight into the spatial dependence structure of the process, informing relationships among measured and unmeasured variables as well as future sampling plans. Zimmerman and Ver Hoef [2024] provide a thorough review of spatial models for ecological and environmental data, providing more detail and justification for the claims made above.

Spatial models also offer many benefits over commonly used machine learning algorithms like random forests [Breiman, 2001, Cutler et al., 2007, James et al., 2017, Kuhn and Silge, 2022]. First, traditional machine learning algorithms do not incorporate spatial dependence and, by consequence, often perform worse than spatial models at characterizing important covariates and predicting at unobserved locations [Fox et al., 2020]. Second, machine learning algorithms struggle to quantify uncertainty in a principled manner, something spatial models excel at. Third, covariate inference is more straightforward using spatial models, as they naturally provide p-values that enable structured testing of hypotheses. Fourth, spatial models provide a rich set of diagnostics like leverage and Cook’s distances [Montgomery et al., 2021], quantities not generally defined for machine learning algorithms. This is not to say that machine learning algorithms cannot be very useful in ecological data analyses, but rather that spatial models are sometimes a more effective tool for modeling spatial data.

Though there are many benefits of spatial models over machine learning algorithms, the reverse is also true. It is much simpler to capture complex nonlinearities using “out-of-the-box” machine learning algorithms, bypassing the tuning often required for spatial models that involve nonlinear components like splines. This can be especially useful for ecological data, which often exhibit nonlinear relationships [Fox et al., 2017]. Machine learning algorithms also typically rely on few, if any, distributional assumptions and can be more robust against unusual observations like outliers [Hastie et al., 2009]. Machine learning algorithms also tend to be much more computationally efficient than spatial models. Importantly, machine learning

algorithms and spatial models can be complementary tools that help shed insight into complicated ecological processes (something we reiterate in the upcoming analyses). An exciting and developing area of statistical research involves incorporating spatial modeling strategies directly into machine learning algorithms [Saha et al., 2023, Heaton et al., 2025].

The first spatially explicit model we define is the spatial linear model. The spatial linear model is a *mixed-effects model* [Pinheiro and Bates, 2000, Bolker, 2015] written as

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots + \beta_p x_{p,i} + \tau_i + \epsilon_i, \quad (1)$$

where $i = 1, 2, \dots, n$ indexes the n observations. The quantity y_i is the response variable (for observation i). The quantity β_j is a slope parameter that controls the average effect of the covariate $x_{j,i}$ on y_i (while holding all other covariates constant). The random error τ_i is a spatial random error, and the random error ϵ_i is the nonspatial (i.e., independent) random error. The spatial random error is spatially structured, capturing spatial residual effects like environmental gradients we do not have data to fully represent. The nonspatial random error is not spatially structured and captures nonspatial residual effects like measurement error and microscale variation. The difference between a traditional (i.e., nonspatial) linear model and a spatial linear model is the inclusion of τ_i (in the spatial linear model). The spatial random error implied by τ_i is governed by a *spatial covariance function*. The spatial covariance function allows the model to incorporate spatial dependence, yielding the improved model performance previously described. Figure 12 shows three different spatial covariance functions that describe different ways the spatial covariance decays with distance. Model parameters and standard errors are estimated from data using likelihood-based [Patterson and Thompson, 1971, Harville, 1977, Wolfinger et al., 1994] or semivariogram-based [Cressie, 1985, Curriero and Lele, 1999] estimation methods. Often, the spatial linear model in Equation 1 is written more compactly using *matrix notation*:

$$\mathbf{y} = \mathbf{X}\beta + \tau + \epsilon, \quad (2)$$

where $\mathbf{X}$ is a matrix that holds the j covariates for each of n observations, and $\mathbf{y}$, β , τ , and ϵ are vectors that contain each of the n or j quantities from Equation 1. The spatial linear model can be adjusted to account for nonspatial random effects just as one would add random effects to a nonspatial linear mixed model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \tau + \epsilon, \quad (3)$$

where $\mathbf{Z}$ is a random effect design matrix (dimension $n \times k$) that indexes each of the k random effects in $\mathbf{u}$. For more information about specific forms of various spatial covariance functions, see Schabenberger and Gotway [2017] and Zimmerman and Ver Hoef [2024].

Like the traditional linear model, the spatial linear model can capture complex nonlinearities in the data through the use of interaction effects [Faraway, 2002, Lenth and Piaskowski, 2025], splines [Chambers and Hastie, 1992, Ruppert et al., 2003], additive models [Wood, 2017, Pedersen et al., 2019], and penalized regression models like the ridge and lasso [Tibshirani, 1996].

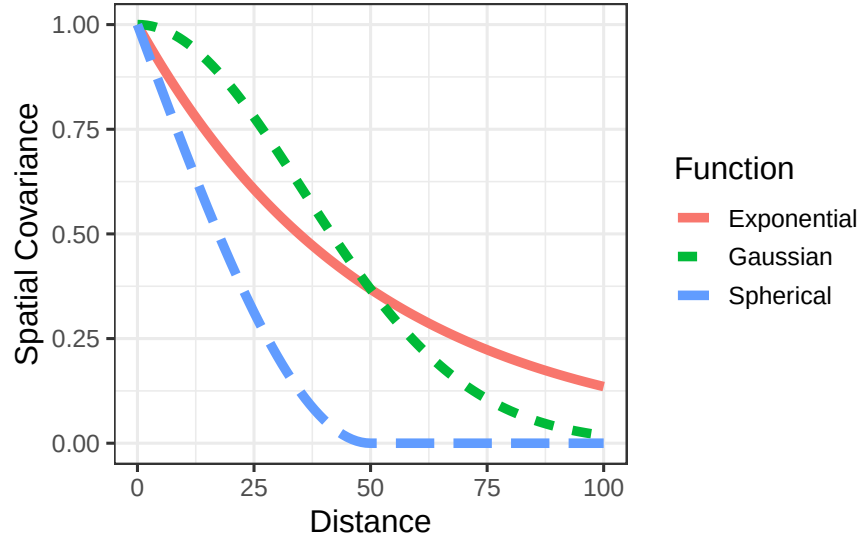


Figure 12: Three spatial covariance functions that represent different spatial dependence behavior.

Generally, this is accomplished by leveraging *basis functions* that break down nonlinear behavior into a set of additive linear components that may include penalty terms which guard against overfitting. Furthermore, there is no assumption that y_i itself is normally distributed; rather, distributional assumptions are placed on the random errors after incorporating the slope parameters. This clarification implies the spatial (and nonspatial) linear models can be quite effective at modeling a range of data that are not normally distributed. For highly nonnormal data, tools like generalized linear models, which we discuss later, can be helpful. Finally, under general conditions, the slope parameters of the spatial (and nonspatial) linear models are approximately normally distributed with a large sample size (see [Casella and Berger \[2024\]](#) for a general justification via the Central Limit Theorem and [Zhang and Zimmerman \[2005\]](#) for nuances regarding spatial data). This is beneficial because it justifies valid hypothesis testing on the slope coefficients (often of primary ecological interest), no matter the distribution of the data (given certain assumptions are satisfied).

One of the most useful applications of the spatial linear model is predicting the response variable at unobserved locations. A very useful spatial prediction method called Kriging has a rich theoretical justification [[Cressie, 1990](#), [Stein, 1999](#)]. Kriging is the spatial analogue to best linear unbiased prediction (BLUP) for mixed models [[Henderson, 1975](#)]. Via Kriging, the spatial linear model produces BLUPs at each unobserved location of interest, alongside prediction intervals to characterize uncertainty. While useful, sometimes the goal is to create a BLUP for an entire region of interest based on realized (i.e., observed) covariate values. This technique, which builds upon the theoretical foundations of Kriging, is called block Kriging [[Ver Hoef, 2002](#), [Cressie, 2006](#)]. The intuition is to make point predictions at a very dense grid in the

Table 11: Common generalized linear model families, link functions, and appropriate data types.

Family	Link Function	Link Name	Data Type
Binomial	$f(\mu) = \log(\mu/(1 - \mu))$	Logit	Binary; Binary Count
Poisson	$f(\mu) = \log(\mu)$	Log	Count
Negative Binomial	$f(\mu) = \log(\mu)$	Log	Count
Beta	$f(\mu) = \log(\mu/(1 - \mu))$	Logit	Proportion
Gamma	$f(\mu) = \log(\mu)$	Log	Skewed
Inverse Gaussian	$f(\mu) = \log(\mu)$	Log	Skewed

study region and then average these predictions, appropriately characterizing uncertainty. Together, Kriging and block Kriging provide a complete set of tools for spatial prediction, given the fit of a spatial linear model.

Spatial generalized linear models are an extension of spatial linear models to highly nonnormal data [Christensen and Waagepetersen, 2002, Zhang, 2002, Diggle and Ribeiro Jr, 2007, Bonat and Ribeiro Jr, 2016]. Ver Hoef et al. [2024] proposed a novel application of the Laplace approximation to enable restricted maximum likelihood estimation of spatial generalized linear models. This model formulation builds upon Equation 1 and is given by

$$g(\mu_i) = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots + \beta_p x_{p,i} + \tau_i + \epsilon_i, \quad (4)$$

where $g(\mu_i)$ is a “link function” that links the distribution of y_i to a linear function of its mean, μ_i . Common link functions include the log link for count and skewed data and the logit link for binomial and proportion data (Table 11). Using the same argument as for spatial linear models, nonspatial random effects can be added to spatial generalized linear models.

Spatial models are complicated, both theoretically and computationally. This has contributed to their relative underuse in the ecological community. However, recent advances in open-source software have made spatial models more accessible to ecologists who have strong subject matter expertise but may lack sufficient statistical background or computational skills to implement them from scratch. One such advancement is the **spmodel** **R** package [Dumelle et al., 2023], which emulates base-**R** functions like `lm()` and `glm()` to account for spatial dependence via `splm()` and `spglm()`, respectively. Familiar **R** functions like `summary()` and `predict()` can be used directly on models fit via **spmodel**. Importantly, **spmodel** also has tools for spatial random forest residual modeling [Fox et al., 2020], applications to large data of several thousand observations via spatial indexing [Ver Hoef et al., 2023], and cross validation [Stone, 1974, Roberts et al., 2017].